

Universal Error-Reducing Methodology on Option Pricing

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Advisor: Dai Tian-Shyr, Chiu Chun-Yuan

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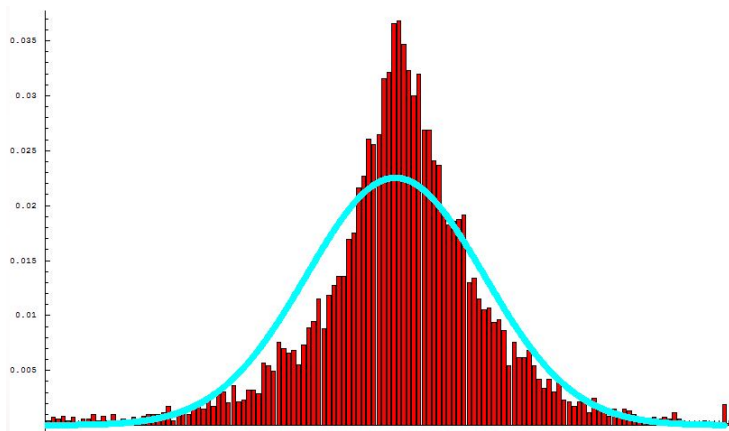
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- Why we use Lévy processes for option pricing?
- What is a Lévy process?
- Option pricing with Lévy process – using FFT
- Our improvements on P.Carr and D.B.Madan (1999). Option valuation using the fast Fourier transform. *Journal of Computational Finance*.

Why we use Lévy processes for option pricing?



What is a Lévy process?

Recall

A stochastic process $X = \{X_t, t \geq 0\}$ is a *standard Brownian motion* on some probability space $(\Omega, \mathcal{F}, \mathbb{P})$ if

- (i) $X_0 = 0$ a.s.
- (ii) X has independent increments. (i.e. $X_{t_2} - X_{t_1} \perp\!\!\!\perp X_{t_4} - X_{t_3}$)
- (iii) X has stationary increments. (i.e. $X_{t_1+s} - X_{t_1} \stackrel{d}{=} X_{t_2+s} - X_{t_2}$)
- (iv) $X_{t+s} - X_t \sim N(0, s)$
- (i.e. $X_{t+s} - X_t$ has characteristic function $\phi(u) = e^{-\frac{1}{2}s^2 u^2}$)

If we generalize ϕ as

$$\phi(u) = e^{-\frac{1}{2}s^2 u^2 + i\gamma u + \int_{-\infty}^{\infty} (e^{iux} - 1 - iux \mathbf{1}_{\{|x| \leq 1\}}) \nu(dx)}$$

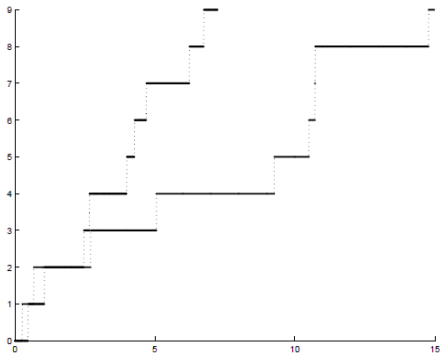
then X defines a Lévy process.

What is a Lévy process?

Example (Poisson process)

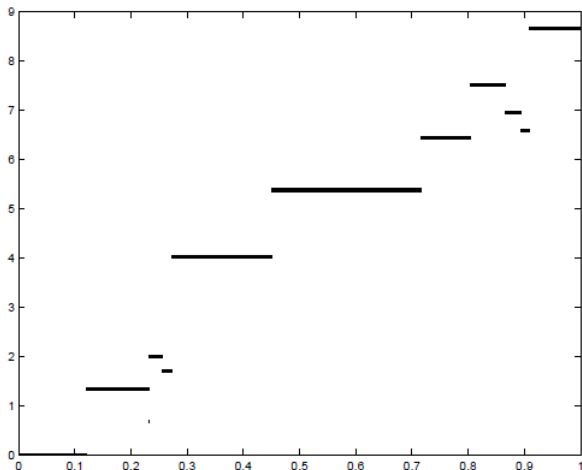
A Poisson process is a stochastic process such that the number of events in time interval $(t, t + s]$ follows a Poisson distribution with associated parameter λs .

$$\text{(i.e. } \mathbb{P}(N(t + s) - N(t) = k) = \frac{e^{-\lambda s}(\lambda s)^k}{k!} \text{ } k = 0, 1, 2, \dots)$$



What is a Lévy process?

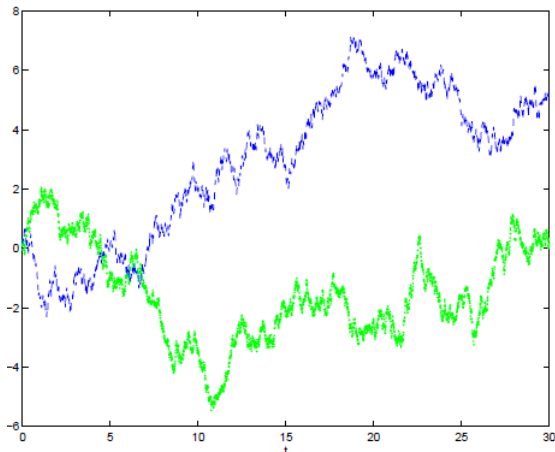
Example (Compound Poisson process(with normal jump size))



What is a Lévy process?

Example (standard Brownian motion)

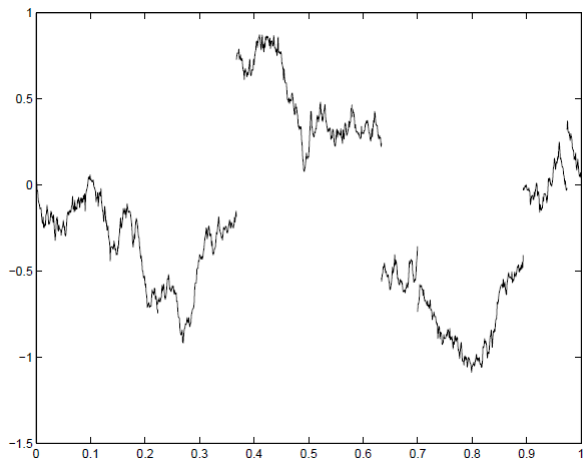
$$d\log S = \mu dt + \sigma dW$$



What is a Lévy process?

Example (Merton's jump diffusion model)

$$d\log S = \mu dt + \sigma dW + J dN, \quad J \sim N(\alpha, \beta)$$



Option pricing with Lévy process – using FFT

The FFT is an efficient algorithm for computing the sum

$$w(k) = \sum_{j=1}^N e^{-i\frac{2\pi}{N}(j-1)(k-1)} \chi(j) \quad \text{for } k = 1 \dots N$$

Carr and Madan(1999) shows that

$$C_T(k) = \frac{e^{-\alpha k}}{2\pi} \int_{-\infty}^{\infty} e^{-ivk} \psi_T(v) dv = \frac{e^{-\alpha k}}{\pi} \int_0^{\infty} e^{-ivk} \psi_T(v) dv$$

where $\psi_T(v) = \frac{e^{-rT} \phi_T(v - (\alpha+1)i)}{\alpha^2 + \alpha - v^2 + i(2\alpha+1)v}$ is a function of a characteristic function.

$$\begin{aligned} \therefore C_T(k) &\approx \frac{e^{-\alpha k}}{\pi} \sum_{j=1}^N e^{-iv_j k} \psi_T(v_j) \eta \\ &= \frac{e^{-\alpha k}}{\pi} \sum_{j=1}^N e^{-i\frac{2\pi}{N}(j-1)(k-1)} e^{ibv_j} \psi_T(v_j) \eta \end{aligned}$$

Our improvements on Carr and Madan (1999)

Our idea: $\psi_T(v) = \psi_T^{proxy}(v) + \psi_T^{residue}(v)$

$$\therefore C_T(k) = \frac{e^{-\alpha k}}{\pi} \int_0^\infty e^{-ivk} \psi_T(v) dv$$

$$= \frac{e^{-\alpha k}}{\pi} \int_0^\infty e^{-ivk} [\psi_T^{proxy}(v) + \psi_T^{residue}(v)] dv$$

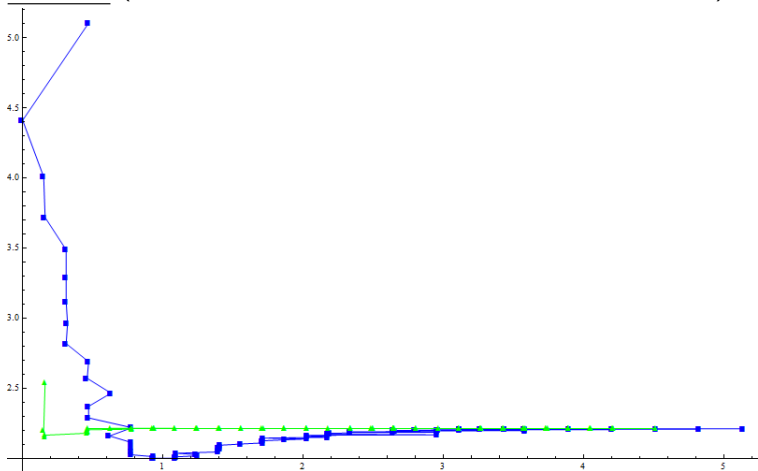
$$= \underbrace{C_T^{proxy}(k)}_{\text{analytic formula}} + \underbrace{\frac{e^{-\alpha k}}{\pi} \int_0^\infty e^{-ivk} \psi_T^{residue}(v) dv}_{FFT}$$

In Merton's jump-diffusion model,

$$C_T^{MJD}(k) = \sum_{j=0}^{\infty} \frac{e^{-\lambda' t} (\lambda' T)^j}{j!} C_T^{BS}(k)(S_0, T, \sigma_j, r_j) := C_T^{proxy}(k)$$

Our improvements on Carr and Madan (1999)

Example (FFT pricing on DE model with MJD proxy used)



Our improvements on Carr and Madan (1999)

Example (FFT pricing on VG model with MJD proxy used)

